

ARPON KAPURIA

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EDUCATION

National Institute of Technology Tiruchirappalli
Bachelor of Technology (Computer Science and Engineering)

Tiruchirappalli, India
November 2020 – June 2024

Kushtia Government College, Kushtia
HSC (Science) | GPA: 5 / 5

Kushtia, Bangladesh
2019

Kushtia Zilla College, Kushtia
SSC (Science) | GPA: 5 / 5

Kushtia, Bangladesh
2017

WORK EXPERIENCE

Advanced Machine Intelligence Research Lab
Research Assistant | *Advisor: Prof. M. Firoz Mridha*

Dhaka, Bangladesh
November 2025 – Present

- Engaged in research on Generative NLP and assisted in implementing deep learning models.

Indian Institute of Technology Bombay
Research Intern | MeDAL Lab | *Advisor: Prof. Amit Sethi*

Mumbai, India
May 2023 – July 2023

- Conducted an in-depth survey on self-supervised learning Frameworks and reproduced BYOL and simCLR in **PyTorch**. Experimented with loss functions and architectural variations to study their impact on learned representations.
- Implemented a DICOM anonymization pipeline using **Pydicom** to remove patient metadata from **5,000+ scans** and integrated CRAFT model to detect text regions in X-ray images for masking PHI.

National Institute of Technology Tiruchirappalli
Undergraduate R&D Assistant | Industrial Automation Lab

Tiruchirappalli, India
October 2022 – February 2023

Project: Assam Smart Home Automation

Advisors: Prof. M. Brindha, Prof. G. S. Ilango

- Developed a cross-platform application to automate energy operations and digital billing for a solar-powered microgrid in Assam, India, serving **20+ households**, as part of a govt. funded project.
- Integrated **Flutter** frontend with Django-based REST API and deployed an MQTT broker enabling real-time data exchange with **50+ IoT sensors** for energy consumption and control.

PROJECTS

Agentic RAG System with Hybrid Retrieval & Scalable Deployment

March 2026 – April 2026

- Built an end-to-end **Agentic RAG pipeline** with persistent conversational memory, supported by a distributed ingestion pipeline using **Ray Data** and event-driven **S3** ingestion to transform raw documents into embeddings and knowledge-graph structures.
- Combined vector search (**Qdrant**) with graph traversal (**Neo4j**) and orchestrated it via **LangGraph** with HyDE-based query rewriting, planner-driven routing, and tool execution for entity-aware, multi-hop reasoning beyond embedding-only retrieval.
- Provisioned infrastructure using **Terraform** on **AWS** (VPC, EKS, S3, IAM) and deployed LLM inference via **Ray Serve** (vLLM backend) with autoscaling, semantic caching, observability (**Prometheus**, **Grafana**, **Opentelemetry**), and evaluation (**RAGAS**, **LLM-as-judge**) to maintain system reliability and answer faithfulness under load.

SKILLS

- Programming Languages:** Python, C, C++, SQL, Bash
- AI / ML:** PyTorch, Hugging Face, LangChain, LangGraph, vLLM, MLflow, DVC
- Backend & Data:** FastAPI, Ray (Data, Serve), PostgreSQL, Redis, Vector DB
- Infrastructure & MLOps:** Git, Docker, Kubernetes, Terraform, AWS (S3, EKS, IAM, VPC), CI/CD
- Observability:** LangSmith, Prometheus, Grafana, OpenTelemetry